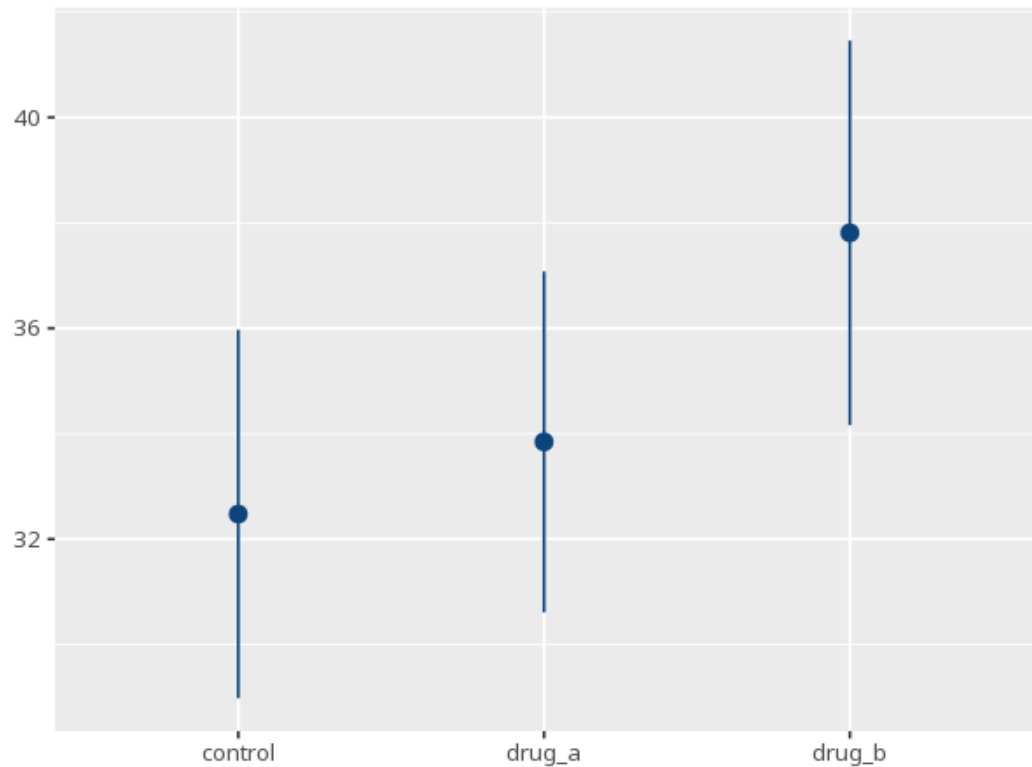


1 Welch's ANOVA

Group	N	Mean	Std. Dev.
control	20	32.48	7.47
drug_a	20	33.84	6.91
drug_b	20	37.81	7.79

F	df1	df2	p	ω^2 [95% CI]
2.58	2	37.90	.089	0.07 [0.00, 1.00]

Pair	Mdiff	padj	95% CI
control - drug_a	-1.37	.820	[-6.92, 4.18]
control - drug_b	-5.33	.082	[-11.22, 0.55]
drug_a - drug_b	-3.96	.218	[-9.65, 1.72]



A one-way ANOVA that relaxes the equal-variance assumption but still assumes roughly normal data within each group. A significant F means at least one group mean differs from the others; the Games-Howell post-hoc (also variance-robust) then shows which specific pairs differ. Your significance threshold (α) is 0.05.

Welch's ANOVA gave $F(2,37.90)=2.58$, $p = .089$, $\omega^2=.07$ [.00, 1.00].

Statistical basis: Welch's ANOVA (heteroscedasticity-robust F, fractional df) (Welch, B. L. (1951). "On the comparison of several mean values: an alternative approach." *Biometrika*, 38(3-4), 330-336.); Games-Howell post-hoc comparisons (also variance-robust) (Kassambara A (2023). *rstatix: Pipe-Friendly Framework for Basic Statistical Tests*. R package.); ω^2 effect size (F-to- ω^2 conversion, no fitted aov model needed) (Ben-Shachar MS, Lüdtke D, Makowski D (2020). "effectsize: Estimation of Effect Size Indices and Standardized Parameters." *Journal of Open Source Software*, 5(56), 2815.).